

CHENLU YE

Computer Science, University of Illinois Urbana-Champaign
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RESEARCH INTERESTS

- Self-improving agents and auto-research;
- Reinforcement learning for agentic LLM post-training;
- Decision-making problems;

EDUCATION

University of Illinois Urbana-Champaign Ph.D. candidate, <i>Computer Science</i> Advisor: Prof. Tong Zhang	Urbana, USA 2024.08 – Present
The Hong Kong University of Science and Technology MPhil, <i>Artificial Intelligence</i> Advisor: Prof. Tong Zhang	Hong Kong, China 2021.09 – 2024.08
University of Science and Technology of China Bachelor of Science, <i>Statistics</i>	Hefei, China 2017.09 – 2021.06

RESEARCH EXPERIENCE

Research Intern, Meta MSL / FAIR Managers: Sainbayar Sukhbaatar and Jason Weston	2026.05 – Present
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- Developed a **meta-optimization framework** that improves agent skills and file-system memory from experience for continual learning and synthetic data generation.
- Developed a skill-learning loop that distills agent reflections into reusable **task-authoring and rubric-design skills**, raising the mean best held-out task-generation pass rate from **5.7% to 40.6% (7.1×)** after 20 optimization iterations.
- Investigating **compute-efficient training for autonomous research agents**, with a focus on sustaining performance over long-horizon, multi-turn research tasks under limited training budgets.

Applied Scientist Intern, Amazon Hosts: Dr. Zhou Yu, Dr. Ziji Zhang	2025.05 – 2026.05
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- Led the development of **Adaptive Layerwise Perturbation (ALP)** to improve off-policy RL robustness to system noise and training–inference mismatch through layerwise policy perturbations. [Paper]
- Demonstrated improvements in training stability, exploration, and task performance on mathematical reasoning and multi-turn tool-integrated reasoning tasks.
- Proposed **Process Consistency Filter (PROF)**, using process–outcome consistency for RL data selection while controlling the correct-to-incorrect response ratio. [Paper]
- Developed PROF-GRPO to improve final-answer accuracy and intermediate reasoning quality; conducted ablations on separately filtering correct and incorrect responses.

RESEARCH EXPERIENCE – CONTINUED

Ph.D. Student, University of Illinois Urbana-Champaign

2024.09 – Present

Advisor: Prof. Tong Zhang

- Developed **Reinforce-Ada**, an adaptive sampling framework that allocates generation budgets across prompts during online RL post-training to preserve informative training signals and increase their diversity. [Paper] [Code]
- Proposed a **self-rewarding correction** framework to improve model self-verification and correction in mathematical reasoning. [Paper]
- Proposed an **online iterative RLHF** framework, implemented it with online DPO, and extended it to general preference models. [Paper] [Code]

MPhil Student, The Hong Kong University of Science and Technology

2021.09 – 2024.08

Advisor: Prof. Tong Zhang

- Formulated RLHF as a reverse-KL-regularized contextual bandit problem under Bradley–Terry preferences and developed statistically efficient algorithms with finite-sample guarantees.
- Developed uncertainty-weighted algorithms for corruption-robust reinforcement learning across online and offline, value-based and model-based settings.

Visiting Research Scholar, University of California, Los Angeles

2023.08 – 2023.12

Host: Prof. Quanquan Gu

- Proposed reinforcement learning algorithms robust to adversarial corruption in online and offline decision-making.

SELECTED PUBLICATIONS AND PREPRINTS

(* denotes equal contribution)

- [1] **Adaptive Layerwise Perturbation: Unifying Off-Policy Corrections for LLM RL**
Chenlu Ye*, Xuanchang Zhang*, Yifan Hao*, Zhou Yu, Ziji Zhang, Abhinav Gullapalli, Hao Chen, Jing Huang, Tong Zhang. [Preprint]
- [2] **Reinforce-Ada: An Adaptive Sampling Framework under Non-linear RL Objectives**
Wei Xiong*, Chenlu Ye*, Baohao Liao*, Hanze Dong*, Xinxing Xu, Christof Monz, Jiang Bian, Nan Jiang, Tong Zhang. [Preprint]
- [3] **Beyond Correctness: Harmonizing Process and Outcome Rewards through RL Training**
Chenlu Ye, Zhou Yu, Ziji Zhang, Hao Chen, Narayanan Sadagopan, Jing Huang, Tong Zhang, Anurag Beniwal. [Preprint]
- [4] **Self-rewarding correction for mathematical reasoning**
Wei Xiong*, Hanning Zhang*, Chenlu Ye*, Lichang Chen, Nan Jiang, Tong Zhang. [Preprint]
- [5] **Online iterative reinforcement learning from human feedback with general preference model**
Chenlu Ye*, Wei Xiong*, Yuheng Zhang*, Hanze Dong*, Nan Jiang, Tong Zhang. [NeurIPS 2024]
- [6] **Iterative preference learning from human feedback: Bridging theory and practice for RLHF under KL-Constraint**
Wei Xiong*, Hanze Dong*, Chenlu Ye*, Ziqi Wang, Han Zhong, Heng Ji, Nan Jiang, Tong Zhang. [ICML 2024]
- [7] **Logarithmic Regret for Online KL-Regularized Reinforcement Learning**
Heyang Zhao*, Chenlu Ye*, Wei Xiong, Quanquan Gu, Tong Zhang. [ICML 2025]

SELECTED PUBLICATIONS AND PREPRINTS – CONTINUED

- [8] **Catoni Contextual Bandits are Robust to Heavy-tailed Rewards**
Chenlu Ye, Yujia Jin, Alekh Agarwal, Tong Zhang. [ICML 2025 Spotlight]
- [9] **Understanding Overadaptation in Supervised Fine-Tuning: The Role of Ensemble Methods**
Yifan Hao*, Xingyuan Pan*, Hanning Zhang*, Chenlu Ye, Rui Pan, Tong Zhang. [ICML 2025]
- [10] **Sharp Analysis for KL-Regularized Contextual Bandits and RLHF**
Heyang Zhao, Chenlu Ye, Quanquan Gu, Tong Zhang. [NeurIPS 2025]
- [11] **Towards robust model-based reinforcement learning against adversarial corruption**
Chenlu Ye*, Jiafan He*, Quanquan Gu, Tong Zhang. [ICML 2024]
- [12] **Corruption-Robust Offline Reinforcement Learning with General Function Approximation**
Chenlu Ye*, Rui Yang*, Quanquan Gu and Tong Zhang. [NeurIPS 2023]
- [13] **Optimal Sample Selection Through Uncertainty Estimation and Its Application in Deep Learning**
Yong Lin*, Chen Liu*, Chenlu Ye*, Qing Lian, Yuan Yao and Tong Zhang. [JMLR 2025]
- [14] **Corruption-Robust Algorithms with Uncertainty Weighting for Nonlinear Contextual Bandits and Markov Decision Processes**
Chenlu Ye, Wei Xiong, Quanquan Gu, and Tong Zhang. [ICML 2023]
- [15] **Provably Efficient High-Dimensional Bandit Learning with Batched Feedbacks**
Jianqing Fan, Zhaoran Wang, Zhuoran Yang, Chenlu Ye (*alphabetical author order*). [Preprint]
- [16] **Daunce: Data Attribution through Uncertainty Estimation**
Xingyuan Pan*, Chenlu Ye*, Joseph Melkonian*, Jiaqi W. Ma, Tong Zhang. [Preprint]
- [17] **Transformers as Multi-task Learners: Decoupling Features in Hidden Markov Models**
Yifan Hao*, Chenlu Ye*, Chi Han, Tong Zhang. [Preprint]

SKILLS

Programming and ML: Python, PyTorch, C++

Developer Tools: Git, Docker

HONORS AND AWARDS

Gold Prize for Outstanding Student Scholarship (1/40)	2020.09
Bronze Prize for Outstanding Student Scholarship	2019.09
Bronze Prize for Outstanding Student Scholarship	2018.09

PROFESSIONAL SERVICE

Conference Reviewer: ICML, NeurIPS, ICLR, AISTATS.

Journal Reviewer: JMLR, Machine Learning, Artificial Intelligence.